Model answer to Q1(a): Textbook and discussion in class

i) What are the characteristics of a Comms link? Briefly explain the importance of latency in real-time communication.

Range, information rate, latency, mobility, installation issues, cost, reliability, power-consumption, size, robustness, lifetime, security are some of the characteristics of a Comms link.

Latency is important in real-time applications, because synchronised communication is important in real-time applications such as video calls and online gaming.

[2]

ii) Briefly explain how does optical communication enable very high-frequency transmission, and what are the advantages of using very high frequency in communication?

Optical communication uses the near-infrared region of the spectrum, which provides very high frequencies for carrier signal. That provides higher band rate, higher bandwidth, multiplexing options as well as making the components more compact.

[2]

iii) Give two examples of commonly used optical frequency windows in optical communication and very briefly explain why some optical frequency ranges are preferable.

The windows centred at 1310 nm and 1550 nm can be given as examples. Window selection is done by considering the attenuation at that window as well as the responsivity of optical components.

[3]

Model answer to Q1(b): Calculated Application of Theory

EM Wave on below plane in Figure 1 is specified as $y = 18\cos[2\pi(5t - 1.8z)]$.

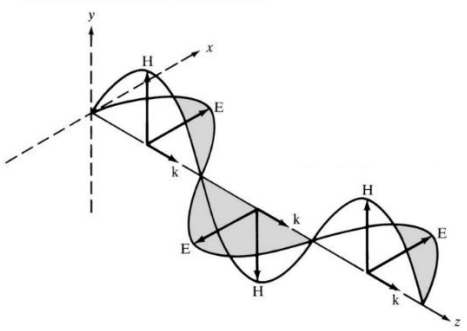


Figure 1.

i) Find the amplitude, the wavelength and the angular frequency of the given EM wave.

General form of an EM wave is $y = A_0 \times \cos(\omega t - kz)$

$$k = \frac{2\pi}{\lambda},\ \omega = 2\pi\nu$$
$$y = A_0[\mu m] \times \cos\left[2\pi\left(\nu t - {}^{z}/_{\lambda}\right)\right]$$

Therefore, on $y = 18\cos[2\pi(5t - 1.8z)]$;

Amplitude = 18 μm.
Wavelength = $\frac{1}{1.8} \approx 555$ nm.
Angular Frequency = $2\pi\nu = 10\pi$ radians.

[3]

ii) Find the displacement of at time $t = 0$ and $z = 7\mu m$.

$$y = A_0 \times \cos\left[2\pi\left(\nu t - {}^{z}/_{\lambda}\right)\right]$$
$$y = 18\cos[2\pi(5t - 1.8z)]$$
$$y = -14.562\ \mu m$$

[3]

Model answer to Q1(c): Calculated Application of Theory

i) A light travelling in air (n_1=1.00) is incident from the side on acetone (n_2=1.356) contained in a glass container (n_3=1.52) with parallel walls. The incoming ray from air to the surface of the glass container makes an angle of $\varphi_1 = 60°$ with respect to the normal, what is angle of refraction φ_2 in the acetone?

$$n_1 \times \sin 60° = n_2 \times \sin \varphi_2$$
$$1 \times \frac{\sqrt{3}}{2} = 1.356 \times \sin \varphi_2$$
$$\varphi_2 = 39.7°$$

[2]

ii) Calculate the critical angle of the interface between a Polymethylmethacrylate (PMMA) surface (n_1=1.489) and water (n_2=1.333).

$$\theta_c = \sin^{-1}\frac{n_2}{n_1}$$
$$\theta_c = \sin^{-1}\frac{1.333}{1.489}$$
$$\theta_c = 63.5°$$

[2]

iii) For a light travelling from air (n_{air}=1.00) into a silica fibre (n_{core}=1.333; $n_{cladding}$=1.30), calculate the numerical aperture.

$$N.A = \sqrt{n_{core}^2 - n_{cladding}^2}$$
$$N.A = \sqrt{1.333^2 - 1.3^2}$$
$$N.A \approx 0.30$$

[2]

Model answer to Q1(d): Textbook and discussion in class

i) How does Rayleigh scattering loss vary with optical frequency?

Rayleigh scattering varies with ${}^{1}/_{\lambda^4}$, so it increases over frequency accordingly ν^4.

[2]

ii) Briefly describe bending losses in optical fibres: where and how they tend to occur, and how they can be prevented.

Bending losses on an optical fibre might be macroscopic or microscopic bending losses. Macroscopic bending occurs while installation and it results in reduce in signal quality due to the signal not to being guided effectively inside the optical channel. To prevent this, ITU-T suggestions, what maximum angles of bending is suggested, should be considered while installation.

Microscopic bending occurs due to material imperfections or mechanical stresses on the optical fibre and similarly it results in reduce in signal quality due to the signal not to being guided effectively inside the optical channel. To prevent this, optical fibre should be fabricated with enhanced methods (less contamination, pure materials) and mechanical stress should not be created on optical fibres.

[3]

iii) A communication link is needed between the Unit A and the Unit B of a factory which are 90 metres apart each other. The transmitter and receiver system can tolerate a maximum link loss of 12 dB with a safety margin 2 dB included. This communication line will need two connectors. As an engineer, you are allocated £600 for this project and asked to choose one of the options below. Based on considering the losses and cost, decide the most suitable option.

Option 1	Option 2
Glass optical fibre, with attenuation of 0.35 dB/km. £2 per metre of glass optical fibre. Connector loss: 0.3 dB/km per connector, £10 each. Installation cost: £200	Plastic optical fibre, with attenuation of 100 dB/km. £0.80 per metre of plastic optical fibre. Connector loss: 0.6 dB/km per connector, £4 each. Installation cost: £80

Total losses and cost are summarised in the table below; choosing option 2 looks preferable, because it is cheaper (£160) and the total loss (10.2 dB) is just acceptable, and under the system tolerance (12 dB).
As the distance is only 90 metre and the system tolerates a maximum link loss of 12 dB, plastic optical fibre can be used in this case.
Approximate answers are also acceptable. Choosing glass optical fibre is also acceptable, as long as the reasoning is justified.

Option 1			**Option 2**		
Material/Service	**Cost (£)**	**Loss (dB)**	**Material/Service**	**Cost (£)**	**Loss (dB)**
Glass Optical Fibre	£180	0.0315	Plastic Fibre	£72	9
Connector	£20	0.6	Connector	£8	1.2
Installation	£200		Installation	£80	
Total (Option 1)	**£400**	**0.6315**	**Total (Option 2)**	**£160**	**10.2**

[7]

iv) If the Unit A is moved to adjacent branch which is 1km away from where the Unit B is, explain if this would affect your choice regarding the signal losses. (Ignore the budget limitation in this case, and you do not need to recalculate the losses numerically, only estimation supported by a brief reasoning is enough.)

Plastic fibre already attenuates the signal by 10.2 dB at that short distance of 90 m. Therefore, if we ignore the budget limitation, glass fibre will be the choice for 1-km long distance, without needing to recalculate the loss. Optical fibre has already 100 dB loss per km.

[3]

v) Silica Fibre A and Silica Fiber B are doped with 7.5% and 15% mole fractions of GeO_2. Estimate the ultraviolet absorptions at 0.9 µm and 1.2 µm for Silica Fiber A and B? From your results, briefly discuss how GeO_2 doping and wavelength affects ultraviolet loss.

Ultraviolet absorptions, $\alpha_{UV(dB/km)}$, at any wavelength given in µm can be calculated empirically as below. mole fraction x of GeO_2

$$\alpha_{UV} = \frac{154.2 \times [mole\ fractions\ of\ GeO_2]}{(46.6 \times [mole\ fractions\ of\ GeO_2]) + 60} \times 10^{-2} \times e^{\frac{4.63}{\lambda}}$$

Therefore,

For 7.5% mole fractions of GeO_2 at 0.9 µm;

$$\alpha_{UV} = \frac{154.2 \times 0.075}{(46.6 \times 0.075) + 60} \times 10^{-2} \times e^{\frac{4.63}{0.9}}$$

$$\alpha_{UV} = 0.31 \text{ dB/km}$$

For 7.5% mole fractions of GeO_2 at 1.2 µm;

$$\alpha_{UV} = \frac{154.2 \times 0.075}{(46.6 \times 0.075) + 60} \times 10^{-2} \times e^{\frac{4.63}{1.2}}$$

$$\alpha_{UV} = 0.086 \text{ dB/km}$$

For 15% mole fractions of GeO_2 at 0.9 µm;

$$\alpha_{UV} = \frac{154.2 \times 0.15}{(46.6 \times 0.15) + 60} \times 10^{-2} \times e^{\frac{4.63}{0.9}}$$

$$\alpha_{UV} = 0.59 \text{ dB/km}$$

For 15% mole fractions of GeO_2 at 1.2 µm;

$$\alpha_{UV} = \frac{154.2 \times 0.15}{(46.6 \times 0.15) + 60} \times 10^{-2} \times e^{\frac{4.63}{1.2}}$$

$$\alpha_{UV} = 0.16 \text{ dB/km}$$

As it can be seen from the results and the empirical formula, ultraviolet absorption loss increases with higher GeO_2 doping; and decreases with increasing wavelength.

[8]

vi) If the scattering loss is estimated as 0.11 dB/km for a silica-based fibre at 1.55 μm, calculate the refractive index of the silica fibre ($T_f = 1400\ K$; $\beta_T = 7 \times 10^{-11}\ {}^{m^2}/_{N}$; $k_B = 1.38 \times 10^{-23}\ {}^{J}/_{K}$; $p = 0.28$).

$$\alpha_{scat[Np/m]} = \frac{\alpha_{scat[dB/km]}}{4.343 \times 1000} = \frac{0.11}{4343} \approx 2.53 \times 10^{-5}$$

$$\alpha_{scat}[Np/m] = \frac{8\pi^3}{3\lambda^4} n^8 p^2 k_B T_f \beta_T$$

$$2.53 \times 10^{-5} = \frac{248.05}{3 \times (1.55 \times 10^{-6})^4} n^8 \times 0.28^2 \times 1.38 \times 10^{-23} \times 1400 \times 7 \times 10^{-11}$$

$$n^{-8} = \frac{248.05}{2.53 \times 10^{-5} \times 3 \times 5.77 \times 10^{-24}} \times 0.28^2 \times 1.38 \times 10^{-23} \times 1400 \times 7 \times 10^{-11}$$

$$n^{-8} = 0.060$$

$$n \approx 1.42$$

[8]

Model answer to Q2(a): Calculated Application of Theory

i) For DWDM system using equal-frequency channel spacing, calculate the wavelength spacing at 1550 nm for ($\Delta\nu$) of 50 GHz, 100 GHz and 200 GHz ($c = 3 \times 10^8$ m/s).

$$c = \lambda\nu$$

$$\nu = \frac{c}{\lambda}$$

$$\frac{\partial\nu}{\partial\lambda} = \frac{\partial}{\partial\lambda}\left(\frac{c}{\lambda}\right)$$

$$\frac{\partial\nu}{\partial\lambda} = -\frac{c}{\lambda^2}$$

$$\partial\nu = -\frac{c}{\lambda^2}\,\partial\lambda$$

$$|\Delta\nu| = \frac{c}{\lambda^2}\,|\Delta\lambda|$$

$$|\Delta\lambda| = \frac{\lambda^2}{c}\,|\Delta\nu|$$

$c = 3 \times 10^8$ m/s

$\lambda = 1550$ nm

At $\Delta\nu = 50$ GHz => $\Delta\lambda = 0.4$ nm

At $\Delta\nu = 100$ GHz => $\Delta\lambda = 0.8$ nm

At $\Delta\nu = 200$ GHz => $\Delta\lambda = 1.6$ nm

[6]

ii) In a DWDM system around 1550 nm with 200 GHz channel spacing, how many independent signal channels could be fit into the bandwidth between 1465 nm and 1625 nm?

Given Bandwidth: $1625 - 1465 = 160$ nm

$$\Delta\lambda = 1.6 \text{ nm at } \Delta\nu = 200 \text{ GHz}$$

$$\frac{160}{1.6} = 100 \text{ independent signal channels.}$$

[6]

iii) In a 16-channel DWDM system using uniform channel spacing is $\Delta\nu = 50$ GHz. In this system, the frequency ν_n correspond to the wavelength λ_n. E.g. $(\lambda_1, \nu_1\ ;\ \lambda_2, \nu_2\ ;\lambda_3, \nu_3\ ; \ldots\ldots ; \lambda_{14}, \nu_{14}\ ; \lambda_{15}, \nu_{15}\ ;\ \lambda_{16}, \nu_{16})$. Assume the wavelength $\lambda_1 = 1535$ nm. Calculate the wavelength spacing between channel 4 and 5 ($\Delta\lambda_a$) and channel 14 and 15 ($\Delta\lambda_b$). From your findings ($\Delta\lambda_a$ and $\Delta\lambda_b$), what can be observed about using equal-frequency channel spacing instead of using equal-wavelength spacing in this wavelength band ($c = 3 \times 10^8$ m/s).

$$\nu_1 = {}^{c}/_{\lambda_1} = 195.439\ THz$$

$$\lambda_2 = {}^{c}/_{\nu_2} = \frac{c}{\nu_1+(2-1)\Delta\nu} = \frac{3\times10^8\ {}^{m}/_{s}}{195.439+0.05\ THz} = 1534.61 \text{ nm}$$

$$\lambda_n = {}^{c}/_{\nu_n} = \frac{c}{\nu_1 + (n-1)\Delta\nu}$$

$$\lambda_4 = {}^{c}/_{\nu_4} = \frac{c}{\nu_1+(4-1)\Delta\nu} = \frac{3\times10^8\ {}^{m}/_{s}}{195.439+3\times0.05\ THz} = 1533.83 \text{ nm}$$

$$\lambda_5 = {}^{c}/_{\nu_5} = \frac{c}{\nu_1+(5-1)\Delta\nu} = \frac{3\times10^8\ {}^{m}/_{s}}{195.439+4\times0.05\ THz} = 1533.44 \text{ nm}$$

$$\lambda_{14} = {}^{c}/_{\nu_{14}} = \frac{c}{\nu_1+(14-1)\Delta\nu} = \frac{3\times10^8\ {}^{m}/_{s}}{195.439+13\times0.05\ THz} = 1529.91 \text{ nm}$$

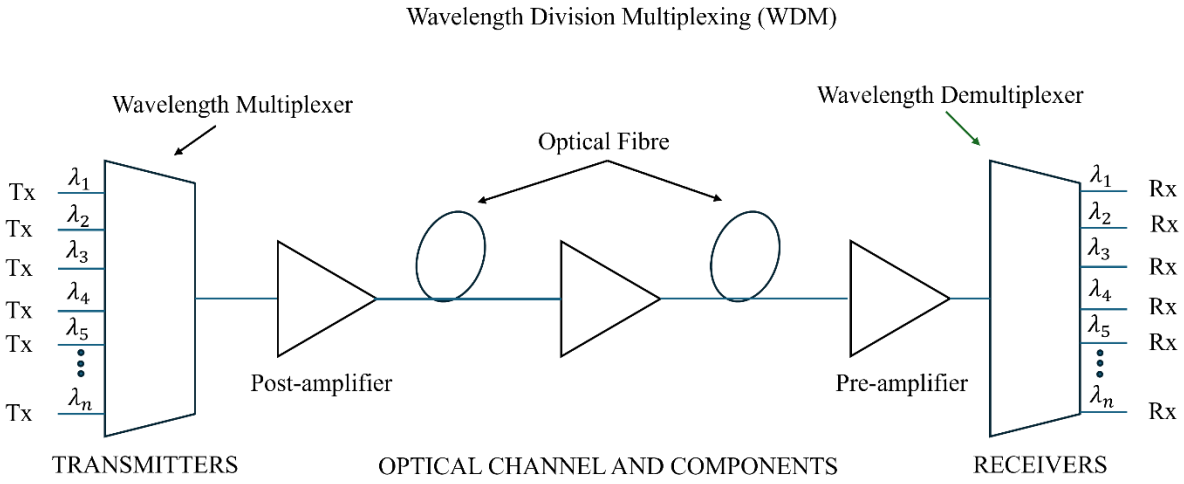
$$\lambda_{15} = {}^{c}/_{\nu_{15}} = \frac{c}{\nu_1+(15-1)\Delta\nu} = \frac{3\times10^8\ {}^{m}/_{s}}{195.439+14\times0.05\ THz} = 1529.53 \text{ nm}$$

Therefore, the wavelength spacing between λ_4 and λ_5; $\Delta\lambda_a$= 0.39 nm and between λ_{14} and λ_{15}; $\Delta\lambda_b$= 0.38 nm

The spacing between different adjacent wavelength $\Delta\lambda$ is changing and not uniform (λ_a and λ_b are not equal); however, the difference is very small and close to an equal-frequency spacing definition at $\Delta\nu = 50$ GHz, which is 0.4 nm.

[10]

iv) Draw the schematic of the Wavelength Division Multiplexing (WDM) and briefly explain the role of the components on the WDM communication line schematic.



Post-amplifiers, in-line amplifiers and pre-amplifiers have similar electronics, they are signal amplifiers. They are named by their place in the communication line. Post-amplifiers are located after the transmitters; in-line amplifiers are located in the middle of the optical link, and pre-amplifiers are located before the receiver block. Wavelength multiplexer and demultiplexer, combines and separates optical signals with different wavelengths in one optical fibre.

[6]

v) Describe briefly how WDM works and list advantages and limitations of WDM Systems.

Advantages can be: efficient use of optical fibre, high bandwidth capacity, scalability, cost is compensated after time.
Disadvantages can be: complex system design, interference between signals, multiple and costly equipment use.

[6]

Model answer to Q2(b): Textbook and discussion in class



Figure 2.

i) Figure 2 above shows a simplified optical communication system. Decide which component/components can be placed at A, B and C from the below list.

- A Laser Diode
- A Light Emitting Diode
- An Erbium-Doped Fiber Amplifier
- An Avalanche Photodiode

'A' can be a Laser Diode or Light Emitting Diode, optical source; electrical signal is converted to optical signal.

'B' can be an Erbium-Doped Fiber Amplifier, optical signal amplifier on the line.

'C' can be Avalanche Photodiode, located at the end of the optical path, converts optical signal to electrical signal.

[6]

ii) A factory needs to monitor temperature and vibration in below factory units to make sure that the environment is safe and suitable for manufacturing and factory operations. For each factory unit given below, briefly explain why optical sensors are better choice than electrical sensors.

a) Power unit that is near high-voltage equipment.

b) Inside a chemical plant where close to explosive gases pipes.

c) Loading/transportation unit with cranes using strong electromagnets.

d) Extreme temperature area housing industrial furnaces.

e) Wastewater treatment unit, which is an area with high humidity and significant exposure to industrial water.

a) No Electromagnetic interference (EMI), electrical insulation and safety (no sparks)

b) No risk of electrical sparks, safer, and mostly more robust to harsh chemical environment.

c) Not affected by EMI, more reliable operation with more accurate results.

d) Optical sensors are immune to high temperatures with a proper coating. Electrical sensors are less accurate due to sensor drift and degradation.

e) Optical sensors are more durable against high humidity and corrosion in wet environments.

[10]

Model answer to Q3(a): Calculated Application of Theory

Power is transmitted at 2.45 GHz.

$$wavelength, \lambda = \frac{c}{f} = 12.245 \text{ cm}$$

i) Calculate the Reactive Near Field distance for a theoretical point source radiator.

$$distance = \frac{\lambda}{2\pi} = 1.95 \text{ cm}$$

[2]

ii) Calculate the Radiative Near Field distance for a theoretical point source radiator.

$$distance = \lambda = 12.245 \text{ cm}$$

[2]

iii) Calculate the Far Field distance for a theoretical point source radiator.

$$distance = \lambda 2 = 24.49 \text{ cm}$$

[2]

iv) Calculate the upper limit of the Rayleigh region for a quarter-wave monopole. What is another name for this region?

This is also known as the reactive near-field and has an upper limit

$$distance = 0.62\sqrt{\frac{(\lambda/4)^3}{\lambda}} = 0.95 \text{ cm}$$

[3]

v) Calculate the upper limit of the Fresnel region for a quarter-wave monopole. What is another name for this region?

This is also known as the radiative near-field and has an upper limit

$$distance = \frac{2(\lambda/4)^2}{\lambda} = 1.53 \text{ cm}$$

[3]

vi) Calculate the lower limit of the Fraunhofer region for a quarter-wave monopole. What is another name for this region?

This is also known as the far field and has an upper limit

$$distance > \frac{2(\lambda/4)^2}{\lambda} = 1.53 \text{ cm}$$

[3]

Model answer to Q3(b): Textbook and discussion in class

Antennas can be polarized in different ways.

i) Very briefly, explain what is meant by a vertically polarized antenna and what kind of carrier modulation was it traditionally used for?

A vertically polarized antenna has its electric field component orientated vertically. This was traditionally used in frequency modulated carrier applications. For example, a quarter-wave antenna used in communications with moving vehicles.

[4]

ii) Very briefly, explain what is meant by a vertically horizontally antenna and what kind of carrier modulation was it traditionally used for?

A horizontally polarized antenna has its electric field component orientated horizontally. This was traditionally used in amplitude modulated carrier applications. For example, a half-wave antenna used in communications with stationary transmitters and receivers.

[4]

iii) What are the linear polarization mismatch losses, in dB, for 45° and 90° mismatch angles between the transmit and receive antennas.

$$-10log\{cos^2\theta\} = 3 \text{ dB and } \infty \text{ dB}$$

[3]

iv) Why do Wi-Fi routers employ antenna diversity switches?

The switch is used to select either a vertically or horizontally polarized antenna, to get the best signal strength.

[2]

v) Very briefly, describe circular polarization and give one application in which it is used.

Circular polarization is used in applications when the exact polarization of the propagating wave cannot be determined. For example, in Low-Earth Orbit communications, when the transmitting antenna on a satellite has a time-varying polarization mismatch, relative the receiving antenna.

[4]

vi) Given a left-hand circularly polarized (LHCP) or anti/counter-clockwise polarized transmit antenna, what kind of polarization is required by the receiving antenna for minimal loss?

There must be a polarization match and so both the transmit and receive antennas must be LHCP for minimal loss.

[2]

vii) Given a right-hand circularly polarized (RHCP) or clockwise polarized transmit antenna, what is the theoretical minimal loss is the receiving antenna is linearly polarized?

There is a minimum polarization mismatch loss of 3 dB.

[2]

Model answer to Q3(c): Calculated Application of Theory

Consider a space-to-Earth wireless power transfer system, whereby a satellite's solar panel array is used to transmit microwave energy at 5.8 GHz down to Earth via a circular parabolic reflecting antenna (Dish). With reference to Figure 1, if the geostationary satellite is positioned 36,000 km directly above the Earth's 3 km diameter circular receiving rectenna array, calculate the following and include all the units:

i) The angle subtended by Earth's receiving rectenna array (shown as a red circle in Figure 1) from the satellite's transmitting antenna ψ.

$$\psi = 2tan^{-1}(1.5/36000) = 83.333x10^{-6} \text{ rads}$$

[2]

ii) The optimum beam solid angle Ω_M for the satellite's antenna. *Hint: for large aperture antennas having a single pencil beam radiation pattern, this is given by $\Omega_M \approx \psi_\theta \psi_\phi$, where ψ_θ and ψ_ϕ are the -3 dB beam widths in the E- and H-planes, respectively.* Clearly state any simplifying assumptions.

$$\Omega \approx \psi_\theta \psi_\phi \text{ assumed} = \psi^2 = 6.944x10^{-9} \text{ sr}$$

[2]

iii) The directivity of the satellite's antenna D_{TX}.

$$\text{Directivity, } D_{TX} = \frac{4\pi}{\Omega} = 1.80956x10^9 = 92.6 \text{ dBi}$$

[2]

iv) The ideal effective aperture area of the satellite's antenna A_{TX}.

$$\text{Free-space wavelength, } \lambda_o = 5.17\ cm$$

$$D_{TX} = \frac{4\pi}{\lambda_o{}^2} A_{TX} \ \therefore A_{TX} = 385.26x10^3 \text{ m}^2$$

[2]

v) The ideal diameter of the satellite's antenna d_{TX}.

$$A_{TX} = \pi \left(\frac{d_{TX}}{2}\right)^2 \ \therefore d_{TX} = 700 \text{ m}$$

[2]

vi) What is a typical efficiency for dish antennas and use your value to calculate the gain of antenna.

$$\text{Assuming an efficiency of } \eta = 60\%, G_{TX} = D_{TX} \cdot \eta = 1.0857 \times 10^9 = 90.4 \text{ dBi}$$

[2]

vii) Is the size of the dish practical. If not, suggest a possible solution.

The dish a way too large to be practical. One solution could be to replace the dish antenna with an active electronically scanned array (AESA) on the backside of the solar panels. This, however, presents other engineering challenges of size, weight and cost.

[2]

Model answer to Q4(a): Calculated Application of Theory

Consider millimetre-wave and sub-terahertz radiometric imaging.

i) State the mathematical expression for Planck's law in terms of zero-point energy, photon energy, thermally-induced energy and emissivity, and include all the units.

$$Average\ Noise\ Energy\ from\ Radiator,\ U = \epsilon \cdot \left(\frac{hf}{2} + \frac{hf}{e^{\frac{hf}{kT}} - 1}\right) \text{ [J]}$$

$$\frac{hf}{2} = \text{ Zero-Point Energy [J]}$$

$$hf = Photon\ Energy \text{ [J]}$$

$$Planck's\ Constant,\ h = 6.626 \times 10^{-34} \text{ [J·s]}$$

$$f = Photon\ Frequency \text{ [Hz]}$$

$$kT = Thermally - induced\ Energy\ for\ a\ charged\ carrier\ (e.g.\ an\ electron)$$

$$Boltzmann's\ Constant,\ k = 1.381 \times 10^{-23}\ [\text{J/K}]$$

$$T = Blackbody\ Temperature\ [\text{K}]$$

[5]

ii) From Plank's law, state the mathematical expression for the average noise power radiated and include all the units.

$$Average\ Noise\ Power\ from\ Radiator,\ P = UB_{pred}\ [\text{W}]$$

$$B_{pred} = Pre - detection\ Bandwidth\ [\text{Hz}]$$

[2]

iii) What is emissivity and its practical implication?

Emissivity represents the efficiency for a body to radiate electromagnetic energy. An ideal 'blackbody' radiator has a 100% efficiency in absorbing and re-radiating electromagnetic energy (with emissivity $\epsilon = 1$).

[3]

iv) What assumption is needed to reduce Planck's law to the Rayleigh-Jeans law?

With low frequencies and/or high temperatures, an approximation can be made (e.g. millimetre-wave radiometers looking at a normal human body gives $hf/kT \cong 0.0465$ at 300 GHz for an average adult human body temperature of 36.55°C or $T = 309.7$ K), giving the Rayleigh-Jeans law:

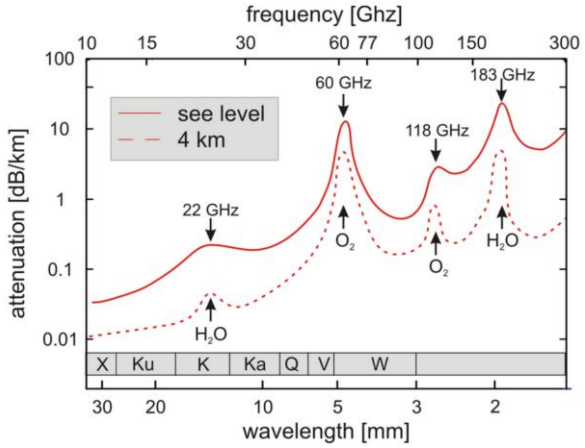
$$e^{\frac{hf}{kT}} = 1 + \frac{1}{1!}\frac{hf}{kT} + \frac{1}{2!}\left(\frac{hf}{kT}\right)^2 + \frac{1}{3!}\left(\frac{hf}{kT}\right)^3 \ldots \approx 1 + \frac{hf}{kT}$$

$$\therefore P \approx kTB_{pred}\ [\text{W}] \text{ at sub-THz frequencies}$$

[4]

v) What are the considerations in choosing the most appropriate frequency bands of operation?

The sub-1 THz band (i.e., frequencies below 1 THz) offers the advantage of unprecedented penetration depth, for seeing through materials such as fabric, ceramic, plastic, leather and cardboard. Above *ca.* 600 GHz, electromagnetic scattering off woven fabrics is a major limitation, as well as reduced penetration depth. In addition, as a general rule of thumb, the cost of implementing technologies increase with increasing frequency, up to the infrared spectrum. For this reason, moving down the frequency spectrum gives improved penetration depth and is potentially cheaper, but at the expense of spatial resolution. The millimetre-wave (MMW) spectrum (30 GHz to 300 GHz, corresponding to wavelengths in free-space from 10 mm down to 1 mm) offers a compromise in terms of depth of penetration against spatial resolution; technologies at these frequencies will be the focus of this report. Operating at millimeter-wave frequencies has a number of other important issues for their practical implementation. The first is that Planck's law shows that the thermal 'brightness' of an object diminishes as frequency decreases, when compared to their peak values located within the IR spectrum. However, in contrast, atmospheric attenuation tends to improve as frequency decreases; although there are spectral peaks that should be avoided, due molecular resonances (oxygen and water absorption bands) – this explains why some millimetre-wave systems operate below *ca.* 40 GHz and between 70 GHz and 100 GHz.



[8]

vi) What is the difference between passive and active radiometric imaging? Briefly explain the hardware considerations between the two. Comment on any health hazards, if any, between the two.

With passive imaging, the target radiates its intrinsic thermal power, without the need for additional sources of power to illuminate the target. The hardware requires very expensive and sophisticated heterodyne radiometer architectures, operating at room temperature, to be able to detect the low levels of thermal power radiated by the target. For this reason, there is no inherent risk of exposing target to harmful levels of power.

With passive imaging, the target is illuminated by additional sources of sinusoidal carrier signals and this extra power is converted to thermal noise power by the target. The hardware requires multiple sources of illumination, with relatively simple direct detection radiometer architectures, operating at cryogenic temperatures, to be able to detect the higher levels of thermal power radiated by the target. The greater the level of irradiation, by the external carrier signal sources, the greater the levels of radiated thermal noise power. However, for this reason, there is a potential heath risk to the target if irradiated with levels that are beyond those recommended by international standards.

[8]

Model answer to Q4(b): Calculated Application of Theory

A radiometer has a pre-detection bandwidth of 20 GHz and a stand-off range of 2 m for detecting human body targets.

i) Calculate the average noise power radiated by target and include the units. State any assumptions for these calculations.

It is assumed that the radiometer operates at sub-THz frequencies and that the target has a body temperature of T = 36.55°C or 309.7 K. As a result, average noise power generated is:

$$P \approx kTB_{pred} = 85.54 \text{ pW or } -70.68 \text{ dBm}$$

[4]

ii) Calculate the received power flux density from the target and include the units.

$$Received\ Power\ Flux\ Density,\ S_{Rx} = \frac{P}{4\pi R^2} = 1.7 \text{ pW/m}^2$$

[2]

iii) The body scanner has a very large antenna, relative to its operational wavelength, deduce the antenna's equivalent noise temperature for the target. State any simplifying assumptions for these calculations.

It is assumed that the solid angle subtended by the target is bigger than that of the antenna's pencil beam. As a result, the antenna's equivalent noise temperature for the target $T_A \approx T =$ 309.7 K.

[2]

iv) The body scanner has 5 dB noise figure, is cooled down to 77 K and has a post-detection integration time of 50 µs. Calculate the radiometric resolution.

$$Radiometer's\ Physical\ Ambient\ Temperature,\ T_0 = 77\ \text{K}$$

$$Radiometer\ Noise\ Figure, NF = 5\ \text{dB}$$

$$Radiometer's\ Noise\ Factor,\ F_{Rx} = 10^{NF/10} = 3.1623$$

$$Radiometer's\ Equivalent\ Noise\ Temperature,\ T_{Rx} = T_0(F_{Rx} - 1) = 166.5\ \text{K}$$

$$T_{system} = T_A + T_{Rx} = 476.2\ \text{K}$$

Radiometric Resolution, Thermal Sensitivity or Temperature Resolution,

$$\Delta T = \frac{T_{system}}{\sqrt{B_{pred}\,\tau_{posd}}} = \frac{476.2}{1{,}000} = 0.5\ \text{K}$$

[12]